

Практикум

E5:archive

Every question in the topic, one after another. No theory, no ladder, no levels: the practicum has all of that. This is for sitting down and working through the material until the moves stop needing thought.

О практикуме

Первая половина темы `calculus.integration_applications`: 39 блоков и 157 баллов, всё, где вопрос — «чему равна первообразная». Разрез поимённый, списки лежат в `blocks` карточки. Вторая половина, «что этот интеграл измеряет», уходит в E6: 46 блоков и 197 баллов. Доли 0.443 и 0.557 посчитаны по баллам, а не подобраны.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e5-integration

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

Archive E5 — techniques of integration

Every question in the topic, one after another. No theory, no ladder, no levels: the practicum has all of that. This is for sitting down and working through the material until the moves stop needing thought.

36 questions, 145 marks, May 2021 — November 2025. The half of `calculus.integration_applications` where the answer is an antiderivative, or the number an antiderivative gives — substitution, parts, partial fractions, reduction formulae, series. What an integral *measures* is E6.

Grouped by the eight techniques of [calculus-integration.yaml](#), in the order of the practicum's ladder. Within a group the questions run from short to long.

How to use this. Read the question, write the answer in the cell, run it. The check answers at once. Solutions are at

the bottom — all of them, in order; use them when you are stuck or when you are done, not in between.

What the checks do. They differentiate. Not one of them integrates — there is no call to `integrate` anywhere in this part of `kit.py`, and a test proves it by reading the source. A check can *recognise* your antiderivative; it cannot produce one. Definite values are added up by adaptive Simpson straight from the integrand.

Two consequences to know before you start.

The constant is free, in any costume. `+ C`, `- 1`, `+ log(A)/2` all differentiate to zero, so all of them pass. Where a point is given and the constant is pinned, the check is told about the point.

Where the paper prints the answer, this notebook asks for the step. "Show that $\int_1^4 x(\ln x)^2 dx = 32(\ln 2)^2 - 16\ln 2 + \frac{15}{4}$ " has nothing to hand in, so the question here is the antiderivative, or the

same integral between different limits.
Those places are marked.

```
import sys
sys.path.append('.')          # from
practicum/calculus to practicum/kit.py
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Rational, sqrt, pi, E, log, Eq

language('en')                 # this
notebook is in English, and so are the checks

a, b, c, g, n, p, q, r, s = symbols('a b c g
n p q r s')  # letters that stay letters
J = Function('J')              # J(m) means
"the integral of f_m", used in section 6
                                # x, y, t, u,
v, A, C, k already come from kit

print('ready; sympy', sp.__version__)
```

1. The antiderivative is in the table

Five questions, 18 marks. Rewrite each term as something the table knows, integrate it, and — if a point on the graph was given — find the constant instead of leaving it as a letter.

1.1 · May 2022 TZ1 Paper 1 Q1 · 5 marks · no calculator

Find the value of $\int_1^9 \left(3 - \frac{5}{\sqrt{x}}\right) dx$. [5]

Hand in the antiderivative as well as the number — the mark scheme buys them separately.

```
q1_1a = ...      # an antiderivative of 3 -  
5/sqrt(x)  
q1_1b = ...      # the value of the definite  
integral  
  
verify_antiderivative('1.1a', q1_1a, 3 -  
5/sqrt(x), domain=(0.5, 9))  
verify_integral('1.1b', q1_1b, 3 - 5/sqrt(x),  
1, 9)
```

1.2 · May 2025 TZ1 Paper 2

Q11(a)(i) · 1 mark · calculator

In a marathon, the time T in hours has probability density function

$$f(t) = \begin{cases} \frac{4}{21} \left(1 - \cos\left(\frac{4\pi}{9}(t - 2.25)\right) \right), & 2.25 \leq t < 4.5; \\ \frac{4}{21} \left(1 + \cos\left(\frac{\pi}{3}(t - 4.5)\right) \right), & 4.5 \leq t \leq 7.5; \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of $\int_{2.25}^{4.5} f(t) dt$. [1]

The answer is exact, which is the point of the question: the cosine contributes nothing over a whole half-period.

```
q1_2 = ...          # the value of the integral

left_piece = Rational(4, 21)*(1 - cos(4*pi*(t
- Rational(9, 4))/9))
verify_integral('1.2', q1_2, left_piece,
Rational(9, 4), Rational(9, 2), var=t)
```

1.3 · May 2022 TZ2 Paper 2 Q2 · 5 marks · calculator

The derivative of a function g is $g'(x) = 3x^2 + 5e^x$, where $x \in \mathbb{R}$. The graph of g passes through $(0, 4)$. Find $g(x)$. [5]

```
q1_3 = ...          # g(x)

verify_antiderivative('1.3', q1_3, 3*x**2 +
5*exp(x), through=(0, 4))
```

1.4 · November 2025 TZ1 Paper 1 Q9(e) · 4 marks · no calculator

The function f has derivative $f'(x) = 3x^2 + 12x - 15$. Given that $f(-2) = 36$, find $f(x)$. [4]

```
q1_4 = ...          # f(x)

verify_antiderivative('1.4', q1_4, 3*x**2 +
12*x - 15, through=(-2, 36))
```

1.5 · May 2021 TZ1 Paper 2 Q12(b) · 3 marks · calculator

The function f has derivative $f'(x) = \frac{1}{x(k-x)}$, where k is a positive constant,

$x \neq 0$ and $x \neq k$. Part (a) has already written this as $\frac{a}{x} + \frac{b}{k-x}$ with $a = b = \frac{1}{k}$.

Hence find an expression for $f(x)$. [3]

No point is given, so the constant stays free. The check runs your answer at three values of k .

```
q1_5 = ...          # f(x), in terms of k

verify_antiderivative('1.5', q1_5, 1/(x*(k -
x)), domain=(0.1, 0.9),
                        params={k: (1, 3, 8)})
```

2. Substitution

Three questions, 17 marks — all of them Paper 1. A function is sitting beside its own derivative, or can be made to. The step the mark scheme pays for is writing the integrand in the new variable, dx included.

2.1 · May 2022 TZ2 Paper 1 Q7 · 6 marks · no calculator

By using the substitution $u = \sec x$ or otherwise, find an expression for

$$\int_0^{\pi/3} \sec^n x \tan x \, dx$$

in terms of n , where n is a non-zero real number. [6]

2.1a The integrand in terms of u . [2]

2.1b The value of the integral. [4]

```

q2_1a = ...      # the integrand in u, after
du has absorbed dx
q2_1b = ...      # the value of the definite
integral, in terms of n

verify_transformed('2.1a', q2_1a,
sec(x)**n*tan(x), sec(x), domain=(0.1, 1.0))
verify_integral('2.1b', q2_1b,
sec(x)**n*tan(x), 0, pi/3, params={n: (2, 3,
5, -1)})

```

2.2 · November 2022 Paper 1

Q4 · 5 marks · no calculator

The derivative of the function f is $f'(x) = \frac{6x}{1+x^2}$. The graph of $y = f(x)$ passes through the point $(1, 5)$. Find an expression for $f(x)$. [5]

```

q2_2 = ...      # f(x)

verify_antiderivative('2.2', q2_2, 6*x/(1 +
x**2), through=(1, 5))

```

2.3 · May 2023 TZ2 Paper 1

Q12(a) · 6 marks · no calculator

By using an appropriate substitution, show that

$$\int \cos \sqrt{x} \, dx = 2\sqrt{x} \sin \sqrt{x} + 2 \cos \sqrt{x} + C.$$

[6]

Printed, so hand in the route. With $t = \sqrt{x}$:

2.3a the integrand in terms of t , including what dx becomes;

2.3b an antiderivative of *that*, still in terms of t .

Substituting $t = \sqrt{x}$ back into 2.3b gives the printed line.

```
q2_3a = ...      # the integrand in t
q2_3b = ...      # an antiderivative of it,
in t

verify_transformed('2.3a', q2_3a,
cos(sqrt(x)), sqrt(x), new=t, domain=(0.2,
4.0))
verify_antiderivative('2.3b', q2_3b,
2*t*cos(t), var=t)
```

3. Integration by parts

Ten questions, 37 marks. The largest technique in the topic, and the only one spread evenly over all three papers. Choose u to be the factor that simplifies when differentiated; everything else follows.

3.1 · November 2023 TZ1 Paper 1 Q8(a) · 6 marks · no calculator

Find $\int x(\ln x)^2 dx$. [6]

The same question is printed in the November 2023 TZ2 paper, word for word. Those two papers are one paper — see the note at the end.

```
q3_1 = ...          # an antiderivative of x*(ln
x)^2

verify_antiderivative('3.1', q3_1,
x*log(x)**2, domain=(0.4, 3.0))
```

3.2 · November 2023 TZ1 Paper 1 Q8(b) · 3 marks · no calculator

Hence show that $\int_1^4 x(\ln x)^2 dx = 32(\ln 2)^2 - 16 \ln 2 + \frac{15}{4}$. [3]

Printed. Do the same thing between different limits instead: find the exact value of $\int_1^2 x(\ln x)^2 dx$.

```
q3_2 = ...          # the exact value between 1
and 2
```

```
verify_integral('3.2', q3_2, x*log(x)**2, 1,
2)
```

3.3 · May 2025 TZ3 Paper 2 Q12(a) · 6 marks · calculator

Find $\int (x^2 - 5)e^x dx$. [6]

```
q3_3 = ...          # an antiderivative
```

```
verify_antiderivative('3.3', q3_3, (x**2 -
5)*exp(x))
```

3.4 · November 2025 TZ1 Paper 2 Q11(a) · 4 marks · calculator

Use integration by parts to find

$$\int \arccos x \, dx. \quad [4]$$

```
q3_4 = ...          # an antiderivative of
arccos(x)

verify_antiderivative('3.4', q3_4, acos(x),
domain=(-0.9, 0.9))
```

3.5 · November 2025 TZ3 Paper 3 Q2(e) · 4 marks · calculator

By using integration by parts, show that

$$\int_0^{1/\sqrt{3}} \arctan x \, dx = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}.$$

[4]

Printed. Hand in the antiderivative of $\arctan x$ instead — that is the whole of the working, and the substitution of limits is arithmetic.

```
q3_5 = ...          # an antiderivative of
arctan(x)

verify_antiderivative('3.5', q3_5, atan(x))
```

3.6 · May 2025 TZ2 Paper 2 Q11(b), and (c)(ii) · 4 marks · calculator

The time T in minutes that a spinning top stays in motion has probability density function

$$f(t) = \begin{cases} kte^{-3t}, & t \geq 0; \\ 0, & \text{otherwise,} \end{cases} \quad k \in \mathbb{Z}^+.$$

The paper shows that $\int_0^a f(t) dt = \frac{k}{9} [1 - (3a + 1)e^{-3a}]$ — printed, so:

3.6a find an antiderivative of te^{-3t} ;

3.6b hence find k , using that the total probability is 1.

```
q3_6a = ...      # an antiderivative of
t*e^(-3t)
q3_6b = ...      # k

verify_antiderivative('3.6a', q3_6a,
t*exp(-3*t), var=t)
# The whole area under a density is 1. Your k
# goes into the integrand, and the
# check adds it up all the way to infinity by
# pushing the limit out.
verify_integral('3.6b', 1, ... if
blank(q3_6b) else q3_6b*t*exp(-3*t), 0, oo,
var=t)
```


**3.7 — 3.10 · May 2023 TZ1
Paper 3 Q1(b)(c)(ii)(d) · 10
marks · calculator**

This investigation is about the family
 $f_n(x) = x^n e^{-x}$, $x \geq 0$.

Part (b) shows that the area under f_1
between 0 and b is $\frac{e^b - b - 1}{e^b}$ — printed,
so:

3.7 find an antiderivative of $x e^{-x}$. [6]

The total area is $A_n = \int_0^\infty f_n(x) dx$, which
part (c)(i) shows is the limit of the
expression above.

3.8 Write down the value of A_1 . [1]

3.9 Use a GDC, with a suitable upper
limit in place of ∞ , to find A_4 . [2]

3.10 And A_5 . [1]

```
q3_7 = ...          # an antiderivative of
x*e^(-x)
q3_8 = ...          # A_1
q3_9 = ...          # A_4
q3_10 = ...         # A_5

verify_antiderivative('3.7', q3_7, x*exp(-x))
verify_integral('3.8', q3_8, x*exp(-x), 0,
oo)
verify_integral('3.9', q3_9, x**4*exp(-x), 0,
oo)
verify_integral('3.10', q3_10, x**5*exp(-x),
0, oo)
```

4. Partial fractions

Four questions, 23 marks. Factorise the denominator, split, integrate the pieces. A repeated factor needs two fractions and they integrate to different kinds of thing.

4.1 · May 2021 TZ2 Paper 1 Q9 · 7 marks · no calculator

By using the substitution $u = \sin x$, find

$$\int \frac{\sin x \cos x}{\sin^2 x - \sin x - 2} dx.$$

[7]

4.1a the integrand in terms of u ; **4.1b** that integrand in partial fractions; **4.1c** the antiderivative, back in terms of x .

```

q4_1a = ...      # the integrand in u
q4_1b = ...      # split into partial
fractions
q4_1c = ...      # the antiderivative in x

raw = sin(x)*cos(x)/(sin(x)**2 - sin(x) - 2)
verify_transformed('4.1a', q4_1a, raw,
sin(x))
check_apart('4.1b', q4_1b, u/(u**2 - u - 2),
var=u)
verify_antiderivative('4.1c', q4_1c, raw,
domain=(0.2, 1.2))

```

4.2 · November 2021 Paper 2

Q10(e) · 7 marks · calculator

Consider $f(x) = \frac{x^2 - x - 12}{2x - 15}$, so that

$$\frac{1}{f(x)} = \frac{2x - 15}{(x + 3)(x - 4)}.$$

4.2a Express $\frac{1}{f(x)}$ in partial fractions. [3]

4.2b Hence find the exact value of $\int_0^3 \frac{1}{f(x)} dx$, as a single logarithm. [4]

```

q4_2a = ...      # the partial fractions
q4_2b = ...      # the exact value

check_apart('4.2a', q4_2a, (2*x - 15)/((x +
3)*(x - 4)))
verify_integral('4.2b', q4_2b, (2*x - 15)/((x
+ 3)*(x - 4)), 0, 3)

```

4.3 · May 2024 TZ1 Paper 1

Q11(d)(e) · 4 marks · no calculator

With $Q(x) = (x + 1)(2x + 1)$:

4.3a Show that $\frac{1}{(x + 1)Q(x)}$ splits into partial fractions. [2]

4.3b Hence find $\int \frac{dx}{(x + 1)^2(2x + 1)}$. [4]

The repeated factor is the whole question: one of the three pieces integrates to a reciprocal, not a logarithm.

```
q4_3a = ...      # the partial fractions of
1/((x+1)^2 (2x+1))
q4_3b = ...      # the antiderivative

check_apart('4.3a', q4_3a, 1/((x + 1)**2*(2*x
+ 1)))
verify_antiderivative('4.3b', q4_3b, 1/((x +
1)**2*(2*x + 1)), domain=(0.2, 3.0))
```

4.4 · May 2025 TZ2 Paper 3

Q1(e) · 5 marks · calculator

Using partial fractions, show that

$$\int \frac{dv}{1-v^2} = \frac{1}{2} \ln \left| \frac{1+v}{1-v} \right| + \frac{1}{2} \ln A,$$

where A is a positive constant. [5]

Printed, but worth writing out: $\frac{1}{2} \ln A$ is the constant of integration, dressed so that exponentiating both sides comes out clean. The check differentiates, so it does not care what the constant is wearing — try $+ C$ and $- 4$ in its place and watch all three pass.

```
q4_4 = ...      # the antiderivative,
constant dressed however you like

verify_antiderivative('4.4', q4_4, 1/(1 -
v**2), var=v, domain=(0.05, 0.9))
```


5. The quotient is f'/f

Three questions, 14 marks — one Paper 3 investigation. The rule is $\int \frac{f'}{f} = \ln|f| + c$, and the difficulty is never the rule: it is getting the quotient into that shape, and then handling a constant that turns from a summand into a factor.

5.1 • May 2024 TZ2 Paper 3 Q1(c) • 2 marks • calculator

The investigation reaches $\frac{f'(x)}{f(x)} = \frac{g'(x)}{g'(x) - g(x)}$ and asks you to integrate both sides, showing that $f(x) = A \exp\left(\int \frac{g'(x)}{g'(x) - g(x)} dx\right)$. [2]

Printed. Do the left-hand side on the paper's own example instead: part (a) uses $f(x) = \frac{1}{(x-2)^2}$. Find $\int \frac{f'(x)}{f(x)} dx$ for that f .


```
q5_1 = ...          # the antiderivative of f'/f
for f(x) = 1/(x-2)^2

verify_antiderivative('5.1', q5_1, -2/(x -
2), domain=(2.4, 6.0))
```

5.2 • May 2024 TZ2 Paper 3

Q1(d) • 5 marks • calculator

Using the result of part (c) with $A = 1$: consider $g(x) = xe^x$. Find $f(x)$ such that f and g satisfy both $(fg)' = fg' + gf'$ and $(fg)' = f'g'$. [5]

```
q5_2 = ...          # f(x)

# If f = exp(∫ ...), then log f is an
# antiderivative of the integrand – and
# with A = 1 the constant is pinned too, so
# the point goes in as well.
verify_antiderivative('5.2', ... if
blank(q5_2) else log(q5_2), x + 1,
                        through=(0, 0))
```

5.3 • May 2024 TZ2 Paper 3

Q1(e) • 7 marks • calculator

The same, with $g(x) = \sin x + \cos x$ on $0 < x < \pi$ and $A = 1$. Give your answer in the form $f(x) = \frac{e^x}{h(x)}$, and hand in $h(x)$. [7]

```
q5_3 = ...          # h(x)

verify_antiderivative('5.3', ... if
blank(q5_3) else x - log(q5_3),
                        Rational(1, 2) -
cot(x)/2, domain=(0.2, 2.9))
```

6. Reduction formula

Three questions, 10 marks — one Paper 1 question. The exponent is a letter, so there is no single antiderivative. What there is, is a way down.

Write your formulae with $\int \cos^m x \, dx$ standing for $\int \cos^m x \, dx$. The check evaluates both sides over a fixed interval for several n ; it never finds an antiderivative, so it cannot hand you the formula, only confirm it.

6.1 · May 2025 TZ1 Paper 1 Q12(a) · 4 marks · no calculator

Consider $f_n(x) = \cos^n x$. By writing $\cos^n x$ as $\cos^{n-1} x \cos x$, show that for $n > 1$

$$\begin{aligned} \int \cos^n x \, dx &= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx \\ &= \cos^{n-1} x \sin x - (n-1) \int \cos^n x \, dx. \end{aligned}$$

[4]

Write the right-hand side. The check will confirm the identity is true — get one sign or one factor of $(n-1)$ wrong and it will not.

```
q6_1 = ...          # the right-hand side, using
J(n) and J(n-2)

verify_reduction('6.1', q6_1, cos(x)**n, n,
J)
```

6.2 · May 2025 TZ1 Paper 1

Q12(b) · 2 marks · no calculator

Hence show that for $n > 1$

$$\int f_n(x) \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int f_{n-2}(x) \, dx.$$

[2]

Two marks for collecting the like integrals onto one side and dividing. The version in 6.1 is unusable as it stands; this one is not.

```
q6_2 = ...          # the collected right-hand
side

verify_reduction('6.2', q6_2, cos(x)**n, n,
J)
```

6.3 · May 2025 TZ1 Paper 1

Q12(c) · 4 marks · no calculator

Hence find an expression for $\int \cos^4 x \, dx$, giving your answer in the form $p \cos^3 x \sin x + q \cos x \sin x + rx + c$ with $p, q, r \in \mathbb{Q}^+$. [4]

Two steps down: $n = 4$, then $n = 2$, then $\int \cos^0 x \, dx = x$.

```
q6_3 = ...          # an antiderivative of
cos^4(x)

verify_antiderivative('6.3', q6_3, cos(x)**4)
```

7. Integrate the series

Five questions, 12 marks. The function has no elementary antiderivative, or has one you are not allowed to write down yet. Expand, integrate term by term, and stop exactly where you were told to.

7.1 · November 2023 TZ1 Paper 1 Q11(d)(iii)(e) · 3 marks · no calculator

Consider $f(x) = e^{\cos 2x}$.

7.1a Write down the Maclaurin series of f up to and including the term in x^2 . [1]

Part (e) uses those two terms to show that $\int_0^{1/10} f(x) dx \approx \frac{149e}{1500}$ — printed, so:

7.1b use the same two terms to approximate $\int_0^{1/2} e^{\cos 2x} dx$. Exact answer. [2]

The same question is printed in the November 2023 TZ2 paper, word for word.

```

q7_1a = ...      # the series up to x^2
q7_1b = ...      # the approximation to the
integral from 0 to 1/2

verify_maclaurin('7.1a', q7_1a,
exp(cos(2*x)), terms=2)
verify_integral('7.1b', q7_1b, q7_1a, 0,
Rational(1, 2))

```

7.2 · November 2025 TZ1 Paper 1 Q10(a)(ii) · 2 marks · no calculator

Given that the first four terms of the binomial expansion of $\frac{1}{1+x^2}$ are $1 - x^2 + x^4 - x^6$, find an approximation for $\int \frac{dx}{1+x^2}$ up to and including the term in x^7 . [2]

```

q7_2 = ...      # the approximation, with
its constant

verify_termwise('7.2', q7_2, 1/(1 + x**2), 8)

```

7.3 · November 2025 TZ1 Paper 1 Q10(b)(ii) · 1 mark · no calculator

Given that $\frac{1}{\sqrt{1-x^2}} = 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + \dots$, find a polynomial expression for $\int \frac{dx}{\sqrt{1-x^2}}$ up to and including the term in x^5 . [1]

```
q7_3 = ...          # the polynomial, with its
                    constant

verify_termwise('7.3', q7_3, 1/sqrt(1 -
x**2), 6)
```

7.4 · November 2025 TZ1 Paper 1 Q10(c) · 4 marks · no calculator

The expression $\frac{25}{48} + \frac{k}{1280}$, with $k \in \mathbb{Z}^+$, can be used to approximate $\arcsin \frac{1}{2}$. Use the result of 7.3 to find k . [4]

The mark that most scripts lose here is the first one: recognising that the integral in 7.3 is $\arcsin x$.


```

q7_4 = ...          # k

# Checked against your own 7.3, evaluated
# between 0 and 1/2 – so the constant
# drops out and nothing about the answer is
# given away.
left = ... if blank(q7_3, q7_4) else
Rational(25, 48) + q7_4/S(1280)
right = 0 if blank(q7_3) else q7_3.subs(x,
Rational(1, 2)) - q7_3.subs(x, 0)
verify_exact('7.4', left, right)

```

7.5 · November 2025 TZ3 Paper 3 Q2(f) · 2 marks · calculator

Determine the value of

$$\int_0^{1/\sqrt{3}} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \right) dx,$$

giving your answer to six decimal places.

[2]

Six decimal places, so the check is told to be strict about them.

```

q7_5 = ...          # the value, to six decimal
places

verify_integral('7.5', q7_5, x - x**3/3 +
x**5/5 - x**7/7, 0, 1/sqrt(3),
tol=1e-6)

```

8. The integral is a condition

Three questions, 14 marks. The value of the integral is what you are given. The antiderivative is a step, and the unknown is a limit, a constant, or nothing at all — sometimes symmetry answers the question outright.

8.1 · May 2023 TZ2 Paper 1 Q4 · 6 marks · no calculator

The region R is bounded by the curve $y = \frac{x}{x^2 + 2}$ (for $x \geq 0$), the x -axis and the line $x = c$. The area of R is $\ln 3$. Find the value of c . [6]

Your answer goes in as the upper limit, and the check integrates up to it.

```
q8_1 = ...          # c
```

```
verify_integral('8.1', log(3), x/(x**2 + 2),  
0, q8_1)
```

8.2 · November 2025 TZ1 Paper 2 Q11(b) · 6 marks · calculator

The random variable X has probability density function

$$f(x) = \begin{cases} 3x \arccos(x^2), & 0 \leq x \leq k; \\ 0, & \text{otherwise.} \end{cases}$$

Part (i) shows that $k^2 \arccos(k^2) - \sqrt{1 - k^4} + \frac{1}{3} = 0$ — printed.

8.2 Hence find the value of k , correct to six significant figures. [6]

This is the one place in the topic where a calculator earns its keep: that equation has no closed-form root. Six significant figures means 0.713 is not an answer, and the check is set to notice.

```
q8_2 = ...          # k, to six significant
figures

verify_integral('8.2', 1, 3*x*acos(x**2), 0,
q8_2, tol=1e-6)
```

8.3 · May 2024 TZ1 Paper 1

Q9(c) · 2 marks · no calculator

f is an odd function, and it is given that

$$\int_0^4 f(|x|) dx = 1.6.$$

Write down the value of

8.3a $\int_{-4}^0 f(x) dx$; [1]

8.3b $\int_{-4}^4 (f(|x|) + f(x)) dx$. [1]

f exists only as a graph here, so there is no integrand to hand the check — these two are compared as numbers. It is the only place in the notebook where that happens.

```
q8_3a = ...      # (a)
q8_3b = ...      # (b)
```

```
check_num('8.3a', q8_3a, 2, 'c144de0e3592')
check_num('8.3b', q8_3b, 2, '3135d2d71bff')
```

Solutions

1.1 · May 2022 TZ1 Paper 1 Q1

Rewrite before integrating: $\frac{5}{\sqrt{x}} = 5x^{-1/2}$,

so

$$\begin{aligned}\int (3 - 5x^{-1/2}) dx &= 3x - \frac{5x^{1/2}}{1/2} \\ &= 3x - 10\sqrt{x}.\end{aligned}$$

Then $[3x - 10\sqrt{x}]_1^9 = (27 - 30) - (3 - 10) = -3 + 7 = 4$. **A1A1A1M1A1**

The mark scheme gives one A1 for the rewrite alone. Both the raising of the index and the dividing by the new index are separately marked.

1.2 · May 2025 TZ1 Paper 2 Q11(a)(i)

Over $2.25 \leq t \leq 4.5$ the cosine runs through exactly one half-period — from $\cos(0)$ down to $\cos \pi$ — so its integral is zero and only the constant survives:

$$\begin{aligned}\int_{2.25}^{4.5} \frac{4}{21} \left(1 - \cos\left(\frac{4\pi}{9}(t - 2.25)\right)\right) dt &= \frac{4}{21} \cdot 2.25 \\ &= \frac{9}{21} \\ &= \frac{3}{7}.\end{aligned}$$

A1

Which is the point of the question: the first half of the race accounts for $\frac{3}{7}$ of the runners, exactly, and you can see it without a calculator.

1.3 • May 2022 TZ2 Paper 2 Q2

$g(x) = x^3 + 5e^x + c$, and $g(0) = 4$ gives $0 + 5 + c = 4$, so $c = -1$:

$$g(x) = x^3 + 5e^x - 1. \quad \text{M1A1A1M1A1}$$

The mark scheme insists the substitution be into an expression that *contains* $+c$ — substituting into $x^3 + 5e^x$ scores M0.

1.4 • November 2025 TZ1 Paper 1 Q9(e)

$f(x) = x^3 + 6x^2 - 15x + c$, and $f(-2) = -8 + 24 + 30 + c = 46 + c = 36$, so $c = -10$:

$$\begin{aligned}
 f(x) &= x^3 \\
 &\quad + 6x^2 \\
 &\quad - 15x \\
 &\quad - 10. \quad \text{M1A1M1A1}
 \end{aligned}$$

1.5 • May 2021 TZ1 Paper 2

Q12(b)

With $f'(x) = \frac{1}{kx} + \frac{1}{k(k-x)}$, the second term integrates with a sign change from the inside function:

$$\begin{aligned}
 f(x) &= \frac{1}{k} \ln|x| - \frac{1}{k} \ln|k-x| + c \\
 &= \frac{1}{k} \ln \left| \frac{x}{k-x} \right| + c.
 \end{aligned}$$

M1A1A1

The $-\frac{1}{k} \ln|k-x|$ is where the marks go: $\frac{d}{dx}(k-x) = -1$, and forgetting it flips the fraction inside the logarithm.

2.1 • May 2022 TZ2 Paper 1 Q7

$u = \sec x$ gives $du = \sec x \tan x \, dx$, and the integrand already contains that factor:

$$\sec^n x \tan x \, dx = \sec^{n-1} x \cdot \underbrace{\sec x \tan x \, dx}_{du} = u^{n-1} du.$$

New limits: $x = 0 \rightarrow u = 1$, $x = \frac{\pi}{3} \rightarrow u = 2$.
So

$$\int_1^2 u^{n-1} du = \left[\frac{u^n}{n} \right]_1^2 = \frac{2^n - 1}{n}.$$

A1M1A1A1M1A1

Method 2 in the mark scheme skips the substitution entirely and integrates by inspection: the derivative of $\sec^n x$ is $n \sec^{n-1} x \cdot \sec x \tan x$, so the antiderivative is $\frac{1}{n} \sec^n x$ — same marks, half the writing.

2.2 • November 2022 Paper 1 Q4

$u = 1 + x^2$, $du = 2x \, dx$, so $\int \frac{6x}{1+x^2} dx = 3 \int \frac{du}{u} = 3 \ln(1 + x^2) + c$. Then $f(1) = 5$ gives $3 \ln 2 + c = 5$, so

$$\begin{aligned}
 f(x) &= 3 \ln(1 + x^2) \\
 &\quad + 5 \\
 &\quad - 3 \ln 2. \quad \mathbf{M1A1A1M1A1}
 \end{aligned}$$

No modulus is needed: $1 + x^2 > 0$ always.

2.3 · May 2023 TZ2 Paper 1

Q12(a)

$t = \sqrt{x}$ gives $x = t^2$ and $dx = 2t dt$, so
 $\int \cos \sqrt{x} dx = \int 2t \cos t dt$. Note that the
substitution *created* the factor $2t$ that
makes the next step possible.

By parts with $u = 2t$, $dv = \cos t dt$:

$$\begin{aligned}
 &\int 2t \cos t \\
 dt &= 2t \sin t - \int 2 \sin t \\
 dt &= 2t \sin t + 2 \cos t + C,
 \end{aligned}$$

and $t = \sqrt{x}$ gives the printed answer.
M1A1A1M1A1A1

3.1 · November 2023 TZ1 Paper 1 Q8(a)

$u = (\ln x)^2$, $dv = x \, dx$:

$$\int x(\ln x)^2 dx = \frac{x^2}{2}(\ln x)^2 - \int x \ln x \, dx.$$

The remaining integral is parts again,
 $u = \ln x$, $dv = x \, dx$: $\int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4}$.
Hence

$$\int x(\ln x)^2 dx = \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2} \ln x + \frac{x^2}{4} + c.$$

M1M1A1M1A1A1

3.2 · November 2023 TZ1 Paper 1 Q8(b)

With F as above, $F(1) = \frac{1}{4}$ and $F(2) = 2(\ln 2)^2 - 2 \ln 2 + 1$, so

$$\begin{aligned} \int_1^2 x(\ln x)^2 dx &= 2(\ln 2)^2 \\ &\quad - 2 \ln 2 \\ &\quad + \frac{3}{4} \\ &\approx 0.1743. \end{aligned}$$

The paper's own limits, 1 to 4, work the same way once you replace $\ln 4$ by $2 \ln 2$ — which is what the second M1 is for.

3.3 · May 2025 TZ3 Paper 2 Q12(a)

$$u = x^2 - 5, \quad dv = e^x dx:$$

$$\begin{aligned} \int (x^2 - 5)e^x dx &= (x^2 - 5)e^x - \int 2xe^x dx \\ &= (x^2 - 5)e^x - (2xe^x - 2e^x), \end{aligned}$$

$$\text{that} \quad \quad \quad \text{is} \quad \quad \quad (x^2 - 2x - 3)e^x + c.$$

M1A1M1A1A1A1

The polynomial dies in two steps because differentiating it twice reaches a constant. That is the whole reason this shape always terminates.

3.4 · November 2025 TZ1 Paper 2 Q11(a)

There is no product until you write one:

$$u = \arccos x, \quad dv = dx, \quad \text{so} \quad v = x \quad \text{and} \quad du = -\frac{dx}{\sqrt{1-x^2}}:$$

$$\int \arccos x \, dx = x \arccos x + \int \frac{x}{\sqrt{1-x^2}} dx = x \arccos x - \sqrt{1-x^2} + c,$$

the last integral by $u = 1 - x^2$.

M1A1M1A1

Method 2 substitutes $x = \cos \theta$ first and integrates by parts afterwards — same four marks.

3.5 · November 2025 TZ3 Paper 3 Q2(e)

$u = \arctan x$, $dv = dx$:

$$\begin{aligned} \int \arctan x \, dx &= x \arctan x - \int \frac{x}{1+x^2} dx \\ &= x \arctan x - \frac{1}{2} \ln(1+x^2) + c. \end{aligned}$$

Between 0 and $\frac{1}{\sqrt{3}}$: $\frac{1}{\sqrt{3}} \cdot \frac{\pi}{6} - \frac{1}{2} \ln \frac{4}{3} = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}$. **A1A1A1A1**

3.6 · May 2025 TZ2 Paper 2

Q11(b), (c)(ii)

$u = t$, $dv = e^{-3t} dt$, so $v = -\frac{1}{3}e^{-3t}$:

$$\begin{aligned} \int te^{-3t} dt &= \\ &= -\frac{t}{3}e^{-3t} \\ &\quad + \frac{1}{3} \int e^{-3t} dt \\ &= -\frac{t}{3}e^{-3t} \\ &\quad - \frac{1}{9}e^{-3t} \\ &\quad + c. \end{aligned}$$

Between 0 and a that is $\frac{1}{9}[1 - (3a + 1)e^{-3a}]$,
and multiplying by k gives the printed
line. **M1A1A1A1**

For k : as $a \rightarrow \infty$ the bracket tends to 1, so
the total probability is $\frac{k}{9} = 1$ and $k = 9$.

3.7 — 3.10 · May 2023 TZ1

Paper 3 Q1

$u = x$, $dv = e^{-x} dx$, $v = -e^{-x}$:

$$\begin{aligned} \int x e^{-x} dx &= \\ &- x e^{-x} \\ &+ \int e^{-x} dx \\ &= -(x+1)e^{-x} + c. \end{aligned}$$

Between 0 and b : $-(b+1)e^{-b} + 1 = \frac{e^b - b - 1}{e^b}$, the printed answer.

A1M1A1A1M1A1

A_1 is the limit of that as $b \rightarrow \infty$, which part (c)(i) finds by l'Hôpital: $A_1 = 1$. **A1**

$A_4 = 24$ and $A_5 = 120$ from a GDC with an upper limit around 20 — the integrand at $x = 20$ is $20^5 e^{-20} \approx 0.0066$, small enough.

M1A1A1

And then $A_n = n!$, which part (f) proves by induction. The whole 25-mark investigation is a derivation of the gamma function without naming it.

4.1 · May 2021 TZ2 Paper 1 Q9

$u = \sin x$, $du = \cos x dx$, so the integrand becomes $\frac{u}{u^2 - u - 2} du = \frac{u}{(u+1)(u-2)} du$.

Partial fractions: $\frac{u}{(u+1)(u-2)} = \frac{A}{u+1} + \frac{B}{u-2}$ gives $A(u-2) + B(u+1) = u$; $u = -1$ gives $A = \frac{1}{3}$, $u = 2$ gives $B = \frac{2}{3}$.

$$\begin{aligned}\int &= \frac{1}{3} \ln |u+1| + \frac{2}{3} \ln |u-2| + C \\ &= \frac{1}{3} \ln |\sin x \\ &\quad + 1| \\ &\quad + \frac{2}{3} \ln |\sin x \\ &\quad - 2| \\ &\quad + C.\end{aligned}$$

A1A1M1M1A1A1A1

Note $\sin x - 2 < 0$ always, which is exactly why the modulus is not optional.

4.2 • November 2021 Paper 2 Q10(e)

$\frac{2x-15}{(x+3)(x-4)} = \frac{A}{x+3} + \frac{B}{x-4}$: $x = -3$ gives $-21 = -7A$, so $A = 3$; $x = 4$ gives $-7 = 7B$, so $B = -1$.

$$\begin{aligned}\int_0^3 \left(\frac{3}{x+3} - \frac{1}{x-4} \right) dx &= [3 \ln |x+3| - \ln |x-4|]_0^3 \\ &= (3 \ln 6 - \ln 1) - (3 \ln 3 - \ln 4).\end{aligned}$$

$3 \ln 6 - 3 \ln 3 = 3 \ln 2$, and $+\ln 4 = 2 \ln 2$, so the total is $5 \ln 2 = \ln 32$.

M1A1A1M1A1A1A1

4.3 · May 2024 TZ1 Paper 1 Q11(d)(e)

$$\frac{1}{(x+1)^2(2x+1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{D}{2x+1}$$
 gives $A = -2$, $B = -1$, $D = 4$, so the split is
$$\frac{4}{2x+1} - \frac{2}{x+1} - \frac{1}{(x+1)^2}.$$

Integrating each piece — and the third one is *not* a logarithm:

$$2 \ln |2x+1| - 2 \ln |x+1| + \frac{1}{x+1} + C.$$

A1A1M1A1

The $\frac{1}{2}$ from the inside function of $2x+1$ cancels against the 4, which is why the first coefficient comes out as 2 and not 4.

4.4 · May 2025 TZ2 Paper 3 Q1(e)

$$\frac{1}{1-v^2} = \frac{1}{(1-v)(1+v)} = \frac{1/2}{1-v} + \frac{1/2}{1+v}, \text{ so}$$

$$\begin{aligned}
\int \frac{dv}{1-v^2} &= \\
&= \frac{1}{2} \ln |1-v| \\
&+ \frac{1}{2} \ln |1+v| \\
&+ c \\
&= \frac{1}{2} \ln \left| \frac{1+v}{1-v} \right| + c.
\end{aligned}$$

Writing $c = \frac{1}{2} \ln A$ is legal because A ranges over all positive numbers as c ranges over $\mathbb{R}$, and it makes part (f) — where both sides get exponentiated — come out without a stray e^{2c} .

M1A1M1A1A1

5.1 • May 2024 TZ2 Paper 3

Q1(c)

For $f(x) = (x-2)^{-2}$: $f'(x) = -2(x-2)^{-3}$, so $\frac{f'}{f} = \frac{-2}{x-2}$ and

$$\int \frac{f'(x)}{f(x)} dx$$

$$=$$

$$- 2 \ln |x$$

$$- 2|$$

$$+ c \quad (= \ln |f(x)| + c) .$$

Which is the general fact the printed part uses: integrating f'/f gives $\ln|f|$, and exponentiating turns the additive constant into the multiplier $A = e^c$. **A1A1**

5.2 · May 2024 TZ2 Paper 3

Q1(d)

$g = xe^x$ gives $g' = (x+1)e^x$ and $g' - g = (x+1)e^x - xe^x = e^x$, so

$$\frac{g'}{g' - g} = \frac{(x+1)e^x}{e^x} = x + 1.$$

Then $\int (x+1)dx = \frac{x^2}{2} + x$ and, with $A = 1$, $f(x) = e^{x+x^2/2}$. **A1M1A1M1A1**

The mark scheme awards A0 if $+c$ survives into the final answer: $A = 1$ was given, and the constant has already been used.

5.3 · May 2024 TZ2 Paper 3

Q1(e)

$g = \sin x + \cos x$ gives $g' = \cos x - \sin x$ and $g' - g = -2 \sin x$, so

$$\frac{g'}{g' - g} = \frac{\cos x - \sin x}{-2 \sin x} = \frac{1}{2} - \frac{1}{2} \cot x.$$

Integrating: $\frac{x}{2} - \frac{1}{2} \ln(\sin x)$ — no modulus needed, since $0 < x < \pi$ was given precisely so that $\sin x > 0$. With $A = 1$,

$$\begin{aligned} f(x) &= \frac{e^{x/2}}{\sqrt{\sin x}} \\ &= \frac{e^x}{e^{x/2} \sqrt{\sin x}}, & h(x) \\ &= e^{x/2} \sqrt{\sin x}. \end{aligned}$$

A1M1A1M1A1A1A1

The last two marks are for the rearrangement into the requested form. Stopping at $e^{x/2}/\sqrt{\sin x}$ is a correct function and an incomplete answer.

6.1 — 6.2 · May 2025 TZ1 Paper 1 Q12(a)(b)

Split off one factor and integrate by parts with $u = \cos^{n-1} x$, $dv = \cos x \, dx$:

$$\int \cos^n x \, dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \sin^2 x \, dx.$$

Now $\sin^2 x = 1 - \cos^2 x$ splits the last integral in two:

$$= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx - (n-1) \int \cos^n x \, dx.$$

M1A1A1A1

The formula is useless in that form — the thing you want is on both sides. Collect it:

$$n \int \cos^n x \, dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx, \text{ hence}$$

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

M1A1

6.3 · May 2025 TZ1 Paper 1 Q12(c)

$$n = 4: \int \cos^4 x = \frac{1}{4} \cos^3 x \sin x + \frac{3}{4} \int \cos^2 x \, dx.$$

$$n = 2: \int \cos^2 x = \frac{1}{2} \cos x \sin x + \frac{1}{2} \int dx = \frac{1}{2} \cos x \sin x + \frac{x}{2}.$$

Substituting back,

$$\begin{aligned} \int \cos^4 x \, dx &= \frac{1}{4} \cos^3 x \sin x \\ &\quad + \frac{3}{8} \cos x \sin x \\ &\quad + \frac{3}{8} x \\ &\quad + c, \end{aligned}$$

so $p = \frac{1}{4}$, $q = r = \frac{3}{8}$. **M1M1A1A1**

The alternative route uses the double-angle formula for $\cos^2 x$ instead of a

second application — same marks, and safer if you distrust your own recursion.

7.1 • November 2023 TZ1 Paper 1 Q11(d)(iii)(e)

$\cos 2x = 1 - 2x^2 + \frac{2}{3}x^4 - \dots$, so $e^{\cos 2x} = e \cdot e^{\cos 2x - 1} = e(1 - 2x^2 + \dots)$, and the first two non-zero terms are $e - 2ex^2$.

$$\begin{aligned}\int_0^{1/2} e(1 - 2x^2)dx &= e \left[x - \frac{2x^3}{3} \right]_0^{1/2} \\ &= e \left(\frac{1}{2} - \frac{1}{12} \right) \\ &= \frac{5e}{12} \\ &\approx 1.133.\end{aligned}$$

M1A1A1

The paper's own limits stop at $\frac{1}{10}$, where the approximation is good to four decimal places. At $\frac{1}{2}$ the true value is 1.108, so the error has grown from 10^{-5} to 0.02 — which is what the x^4 term is for.

7.2 · November 2025 TZ1 Paper 1 Q10(a)(ii)

Integrate the four terms one at a time, raising each index and dividing:

$$\begin{aligned} & \int (1 - x^2 + x^4 - x^6) \, dx \\ &= x \\ &\quad - \frac{x^3}{3} \\ &\quad + \frac{x^5}{5} \\ &\quad - \frac{x^7}{7} \\ &\quad + C. \end{aligned}$$

A2

That is the Maclaurin series of $\arctan x$, and the question is one of the standard ways of deriving it.

7.3 · November 2025 TZ1 Paper 1 Q10(b)(ii)

$$\begin{aligned} & \int \left(1 + \frac{x^2}{2} + \frac{3x^4}{8} \right) dx \\ &= x \\ & \quad + \frac{x^3}{6} \\ & \quad + \frac{3x^5}{40} \\ & \quad + C. \quad \mathbf{A1} \end{aligned}$$

7.4 · November 2025 TZ1 Paper 1 Q10(c)

The integral in 7.3 is $\arcsin x$, so

$$\begin{aligned} \arcsin \frac{1}{2} & \approx \frac{1}{2} \\ & + \frac{1}{6} \cdot \frac{1}{8} \\ & + \frac{3}{40} \cdot \frac{1}{32} \\ &= \frac{1}{2} + \frac{1}{48} + \frac{3}{1280} \\ &= \frac{25}{48} + \frac{3}{1280}, \end{aligned}$$

giving $k = 3$. **A1M1A1A1**

The value is 0.523177 against $\pi/6 = 0.523599$ — good to three decimal places from three terms.

7.5 • November 2025 TZ3 Paper 3 Q2(f)

$$\left[\frac{x^2}{2} - \frac{x^4}{12} + \frac{x^6}{30} - \frac{x^8}{56} \right]_0^{1/\sqrt{3}} = 0.158422. \quad \mathbf{A2}$$

Compare with part (e): $\int_0^{1/\sqrt{3}} \arctan x \, dx = 0.158459$. The four terms of the series get five significant figures, and the whole investigation is about how much better each further term makes it.

8.1 • May 2023 TZ2 Paper 1 Q4

$$\int_0^c \frac{x}{x^2 + 2} dx = \left[\frac{1}{2} \ln(x^2 + 2) \right]_0^c = \frac{1}{2} \ln \frac{c^2 + 2}{2},$$

by inspection or $u = x^2 + 2$.

Set that equal to $\ln 3$: $\ln \frac{c^2 + 2}{2} = 2 \ln 3 = \ln 9$, so $c^2 + 2 = 18$, $c^2 = 16$, and $c = 4$ since $c > 0$. **M1A1M1M1A1A1**

Two of the six marks are for the log laws, and one for finishing: $c^2 = 16$ is not an answer to "find the value of c ".

8.2 • November 2025 TZ1 Paper 2 Q11(b)

$u = x^2$ turns $\int_0^k 3x \arccos(x^2) dx$ into $\frac{3}{2} \int_0^{k^2} \arccos u \, du$, and part (a) has already found that antiderivative:

$$\begin{aligned} \frac{3}{2} \left[u \arccos u - \sqrt{1 - u^2} \right]_0^{k^2} \\ = \frac{3}{2} \left(k^2 \arccos(k^2) - \sqrt{1 - k^4} + 1 \right). \end{aligned}$$

Setting that to 1 and dividing by $\frac{3}{2}$ gives the printed equation. Solving it on a GDC: $k = 0.713250$. **M1M1A1A2**

8.3 • May 2024 TZ1 Paper 1 Q9(c)

f is odd, so $f(-x) = -f(x)$ and the graph on $[-4, 0]$ is the graph on $[0, 4]$ turned upside down. For $x \geq 0$, $f(|x|) = f(x)$, so $\int_0^4 f(x) \, dx = 1.6$ and

$$\int_{-4}^0 f(x) \, dx = - \int_0^4 f(x) \, dx = -1.6.$$

For the second: $f(|x|)$ is *even*, so $\int_{-4}^4 f(|x|)dx = 2 \cdot 1.6 = 3.2$; and f is *odd*, so $\int_{-4}^4 f(x)dx = 0$. The total is 3.2. **A1A1**

Two marks, no antiderivative, and no formula for f — the whole question is the two symmetry facts.

What the topic looks like from above

technique	questions	marks
Integration by parts	10	37
Partial fractions	4	23
The antiderivative is in the table	5	18
Substitution	3	17
The quotient is f'/f	3	14
The integral is a condition	3	14
Integrate the series	5	12
Reduction formula	3	10

Three things worth noticing once you have worked through them all.

Parts is a quarter of the topic and the only technique on every paper. Eighteen marks on Paper 1, fourteen on Paper 2, fourteen on Paper 3.

Eighty-six of the 145 marks are Paper 1. This is the least calculator-flavoured topic in the archive, and the 45% of marks that formally allow one is the most misleading figure in the whole series. The calculator does real work exactly twice: A_4 and A_5 in 3.9–3.10, and $k = 0.713250$ in 8.2.

Nine questions of the 36 print their own answer. *Show that, hence show, by using integration by parts, show.* In every one of those this notebook asks for the step instead — the antiderivative, the substituted integrand, or the same integral between other limits — because a printed answer teaches nothing and checks nothing.

A note on November 2023. The corpus holds that session as two zones, TZ1 and TZ2, and counts its questions twice. They

are the same paper: all twelve questions of Paper 1 and nine of the twelve on Paper 2 are identical word for word, and the only line that differs is the paper's own code, 8823-7106 against 8823-7111. Three blocks of this topic — 3.1, 3.2 and 7.1 — are therefore in the corpus twice and here once. That double count is worth 12 of the 157 marks the corpus reports for E5, which is why this notebook says 145.